Operating Systems Lab - 24 Practice Questions with Proper Solutions

Question 1: Student Information

Solution (Shell Script)

```
#!/bin/bash
read -p "Enter Name: " name
read -p "Enter Age: " age
echo "Name : $name"
echo "Age  : $age"
```

Question 2: Marks Calculator

Solution (Shell Script)

```
#!/bin/bash
read -p "Mark1: " m1
read -p "Mark2: " m2
read -p "Mark3: " m3
read -p "Mark4: " m4
read -p "Mark5: " m5
total=$((m1+m2+m3+m4+m5))
average=$((total/5))
echo "Total = $total"
echo "Average = $average"
```

Question 3: Temperature Converter

Solution (Shell Script)

```
#!/bin/bash
read -p "Enter Celsius: " c
f=$((c*9/5+32))
k=$((c+273))
echo "Fahrenheit = $f"
echo "Kelvin = $k"
```

Question 4: Mini Banking System

Solution (Shell Script)

```
#!/bin/bash
read -p "Opening Balance: " bal
read -p "Deposit: " dep
read -p "Withdraw: " wd
close=$((bal+dep-wd))
echo "Closing Balance = $close"
```

Question 5: Even or Odd

Solution (Shell Script)

```
#!/bin/bash
read -p "Enter Number: " n
if ((n%2==0)); then
    echo "Even Number"
else
    echo "Odd Number"
fi
```

Question 6: Vote Eligibility

Solution (Shell Script)

```
#!/bin/bash
read -p "Age: " age
if [ $age -ge 18 ]; then
    echo "Eligible to Vote"
else
    echo "Not Eligible"
fi
```

Question 7: Largest of Two

Solution (Shell Script)

```
#!/bin/bash
read -p "First: " a
read -p "Second: " b
if [ $a -gt $b ]; then
    echo "$a is Largest"
else
    echo "$b is Largest"
fi
```

Question 8: Largest of Three

Solution (Shell Script)

```
#!/bin/bash
read a; read b; read c
if [ $a -gt $b ] && [ $a -gt $c ]; then
 echo "$a is Largest"
elif [ $b -gt $c ]; then
 echo "$b is Largest"
else
 echo "$c is Largest"
fi
```

Question 9: Grade Calculator

Solution (Shell Script)

```
#!/bin/bash
read -p "Marks: " m
if [ $m -ge 90 ]; then
 echo "Grade A"
elif [ $m -ge 80 ]; then
 echo "Grade B"
elif [ $m -ge 70 ]; then
 echo "Grade C"
else
 echo "Fail"
fi
```

Question 10: Leap Year

Solution (Shell Script)

```
#!/bin/bash
read -p "Year: " year
if ((year%400==0 || (year%4==0 && year%100!=0))); then
 echo "Leap Year"
else
 echo "Not Leap Year"
fi
```

Question 11: Bank Withdrawal

Solution (Shell Script)

```
#!/bin/bash
read -p "Balance: " bal
read -p "Withdraw: " wd
if [ $wd -le $bal ]; then
 echo "Transaction Successful"
else
 echo "Insufficient Balance"
fi
```

Question 12: Print 1 to N (for)

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=1;i<=n;i++))
do
 echo $i
done
```

Question 13: Sum of First N (for)

Solution (Shell Script)

```
#!/bin/bash
read n
sum=0
for((i=1;i<=n;i++))
do
 sum=$((sum+i))
done
echo "Sum = $sum"
```

Question 14: Factorial

Solution (Shell Script)

```
#!/bin/bash
read n
fact=1
for((i=1;i<=n;i++))
do
 fact=$((fact*i))
done
echo "Factorial = $fact"
```

Question 15: Multiplication Table

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=1;i<=10;i++))
do
 echo "$n x $i = $((n*i))"
done
```

Question 16: Squares

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=1;i<=n;i++))
do
 echo "$i -> $((i*i))"
done
```

Question 17: Even Numbers

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=1;i<=n;i++))
do
 if ((i%2==0)); then
  echo $i
 fi
done
```

Question 18: Odd Numbers

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=1;i<=n;i++))
do
 if ((i%2!=0)); then
  echo $i
 fi
done
```

Question 19: Countdown

Solution (Shell Script)

```
#!/bin/bash
read n
for((i=n;i>=1;i--))
do
 echo $i
done
```

Question 20: Print 1 to N (while)

Solution (Shell Script)

```
#!/bin/bash
read n
i=1
while [ $i -le $n ]
do
 echo $i
 i=$((i+1))
done
```

Question 21: Sum of First N (while)

Solution (Shell Script)

```
#!/bin/bash
read n
i=1
sum=0
while [ $i -le $n ]
do
 sum=$((sum+i))
 i=$((i+1))
done
echo "Sum = $sum"
```

Question 22: Reverse Number

Solution (Shell Script)

```
#!/bin/bash
read num
reverse=0
while [ $num -gt 0 ]
do
 digit=$((num%10))
 reverse=$((reverse*10+digit))
 num=$((num/10))
done
echo "Reverse = $reverse"
```

Question 23: Count Digits

Solution (Shell Script)

```
#!/bin/bash
read num
count=0
while [ $num -gt 0 ]
do
 count=$((count+1))
 num=$((num/10))
done
echo "Digits = $count"
```

Question 24: Sum of Digits

Solution (Shell Script)

```
#!/bin/bash
read num
sum=0
while [ $num -gt 0 ]
do
 digit=$((num%10))
 sum=$((sum+digit))
 num=$((num/10))
done
echo "Sum of Digits = $sum"
```

OS Assessment 2 - Questions 25 to 31

Q25 Fibonacci

```
#!/bin/bash
read n
a=0;b=1;i=3
echo -n "$a $b "
while [ $i -le $n ]
do
c=$((a+b))
echo -n "$c "
a=$b
b=$c
i=$((i+1))
done
```

Q26 Product of Digits

```
#!/bin/bash
read num
product=1
while [ $num -gt 0 ]
do
digit=$((num%10))
product=$((product*digit))
num=$((num/10))
done
echo $product
```

Q27 Palindrome

```
#!/bin/bash
read num
original=$num
reverse=0
while [ $num -gt 0 ]
do
digit=$((num%10))
reverse=$((reverse*10+digit))
num=$((num/10))
done
if [ $original -eq $reverse ]; then echo Palindrome; else echo 'Not
Palindrome'; fi
```

Q28 Armstrong

```
#!/bin/bash
read num
original=$num
sum=0
while [ $num -gt 0 ]
do
digit=$((num%10))
cube=$((digit*digit*digit))
sum=$((sum+cube))
```

```
num=$((num/10))
done
if [ $original -eq $sum ]; then echo Armstrong; else echo 'Not
Armstrong'; fi
```

Q29 Prime

```
#!/bin/bash
read num
i=2;flag=0
while [ $i -lt $num ]
do
if ((num%i==0)); then flag=1; break; fi
i=$((i+1))
done
if [ $flag -eq 0 ] && [ $num -gt 1 ]; then echo Prime; else echo 'Not
Prime'; fi
```

Q30 Calculator using case

```
#!/bin/bash
read a; read b; read ch
case $ch in
1) echo $((a+b));;
2) echo $((a-b));;
3) echo $((a*b));;
4) echo $((a/b));;
*) echo Invalid;;
esac
```

Q31 Day using case

```
#!/bin/bash
read day
case $day in
1) echo Monday;;
2) echo Tuesday;;
3) echo Wednesday;;
4) echo Thursday;;
5) echo Friday;;
6) echo Saturday;;
7) echo Sunday;;
*) echo 'Invalid Choice';;
esac
```